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Fe-La-Si (Iron-Lanthanum-Silicon)

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The early work of [1972Bod] on this system was reviewed by [1992Rag] and an isothermal section at 600 °C was presented depicting five ternary compounds. Subsequently, [2001Rag] briefly reviewed the work of [1996Lia] depicting the phase equilibria in alloys furnace-cooled from 900 °C. Recently, [2012Nii] reinvestigated this system and presented three isothermal sections at 1300, 1100 and 900 °C and two vertical sections.

Binary Systems

In the Fe-La system, there are no intermediate phases and the mutual solubility between Fe and La is negligible. In the Fe-Si system [1982Kub], the Fe-based face-centered cubic phase γ is enclosed by a loop. The intermediate phases are: α_2 (*B*2, CsCl-type cubic), α_1 (*D*0$_3$, BiF_3-type cubic), Fe_2Si (stable between 1212 and 1040 °C; hexagonal), Fe_5Si_3 (*D*8$_8$, Mn_5Si_3-type hexagonal), FeSi (*B*20-type cubic), $FeSi_2$ (HT) (*tP*3, *P*4/*mmm*; stable between 1220 and 937 °C; also denoted ζ_α) and $FeSi_2$(LT) (ζ_β; *oC*48, *Cmca*; stable up to 982 °C). The La-Si phase diagram [2001Bul] depicts six intermediate phases: La_5Si_3 (D8$_l$, Cr_5B_3-type tetragonal), La_3Si_2 (*D*5$_a$, Si_2U_3-type tetragonal), La_5Si_4 (Zr_5Si_4-type tetragonal), LaSi (*B*27, FeB-type orthorhombic), α$LaSi_2$ (62.5-64 at.% Si; α$GdSi_2$-type orthorhombic), and β$LaSi_2$ (64.5-67 at.% Si; C_c, $ThSi_2$-type tetragonal).

Ternary Compounds

Confirming the observations of [1996Lia], [2012Nii] found two distinct phases of $La(Fe_{1-x}Si_x)_{13}$, in different ranges of *x* along the isoconcentrate line of 7.1 at.% La. Phase 1 (denoted τ_1 here and by [2012Nii] and as τ_{1a} by [2001Rag]) has (at 900 °C) a composition range of $0.086 \le x \le 0.211$ (8-19.6 at.% Si) with $NaZn_{13}$-type of cubic structure. Phase 2 (denoted τ_2 here and by [2012Nii] and as τ_{1b} by [2001Rag]) has a composition range of $0.251 \le x \le 0.398$ (23.3-37 at.% Si) with $Ce_2Ni_{17}Si_9$-type of tetragonal structure. The other ternary phases are: $LaFe_2Si_2$ (denoted here τ_3; *D*1$_3$, Al_4Ba-type tetragonal), $LaFeSi_2$ (τ_4; $CeNiSi_2$-type orthorhombic), LaFeSi (τ_5; *C*38, Cu_2Sb-type tetragonal), and $LaFe_xSi_{2-x}$ (τ_6; *C*32, AlB_2-type hexagonal) [2012Nii]. It may be noted that the present notations τ_3, τ_4, τ_5 and τ_6 do not correspond to those used by [2001Rag]. Table 1 gives the updated structural data from [2012Nii].

Ternary Phase Equilibria

With starting metals of 99.99% Fe, 99.9% La and 99.999% Si, [2012Nii] arc-melted or levitation-melted under Ar atm about 35 ternary alloys. The samples were annealed at 1300 °C (0.02-3 days), 1100 °C (0.13-21 days) and 900 °C (7-50 days) and quenched in ice-water. Diffusion couples were also prepared, annealed at 900 °C for 12 h and quenched in ice-water. The phase equilibria were studied with scanning electron microscopy, electron probe microanalysis, x-ray powder diffraction and differential scanning calorimetry at a heating rate of 10 °C/min.

Three isothermal sections constructed by [2012Nii] at 1300, 1100 and 900 °C are shown in Fig. 1-3. At 1300 °C (Fig. 1), the Si-rich region was not investigated. In the range shown, the ternary phases τ_1, τ_2 and $LaFe_2Si_2$(τ_3) are present. At 1100 and 900 °C (Fig. 2, 3), the plot covers the entire composition range and depicts all the ternary phases listed above. As discussed above, both τ_1 and τ_2 show an

Table 1 Fe-La-Si crystal structure and lattice parameter data [2012Nii]

Phase	Composition, at.%	Pearson symbol	Space group	Prototype	Lattice parameter, nm
$La(Fe,Si)_{13}$(τ_1)	84.9-73.3Fe, 7.1La, 8-19.6Si	*cF*112	$Fm\bar{3}c$	$NaZn_{13}$	*a* = 1.146(a)
$La(Fe,Si)_{13}$(τ_2)	69.6-55.9Fe, 7.1La, 23.3-37Si	*tI*56	*I*4/*mcm*	$Ce_2Ni_{17}Si_9$	*a* = 0.7978(b) *c* = 1.166
$LaFe_2Si_2$(τ_3)	~40Fe, ~20La, ~40Si	*tI*10	*I*4/*mmm*	Al_4Ba	*a* = 0.4091 *c* = 0.4346
$LaFeSi_2$(τ_4)	25Fe, 25La, 50Si	*oC*16	*Cmcm*	$CeNiSi_2$	*a* = 0.4177 *b* = 1.720 *c* = 0.4050
LaFeSi(τ_5)	33.3Fe, 33.3La, 33.3Si	*tP*6	*P*4/*nmm*	Cu_2Sb	*a* = 0.4104 *c* = 0.7142
$LaFe_xSi_{2-x}$(τ_6)	16.7-10Fe, 33.3La, 50-56.7Si	*hP*3	*P*6/*nmm*	AlB_2	*a* = 0.409(c) *c* = 0.4345

(a) At $x = 0.108$; (b) at $x = 0.268$; and (c) at $x = 0.308$

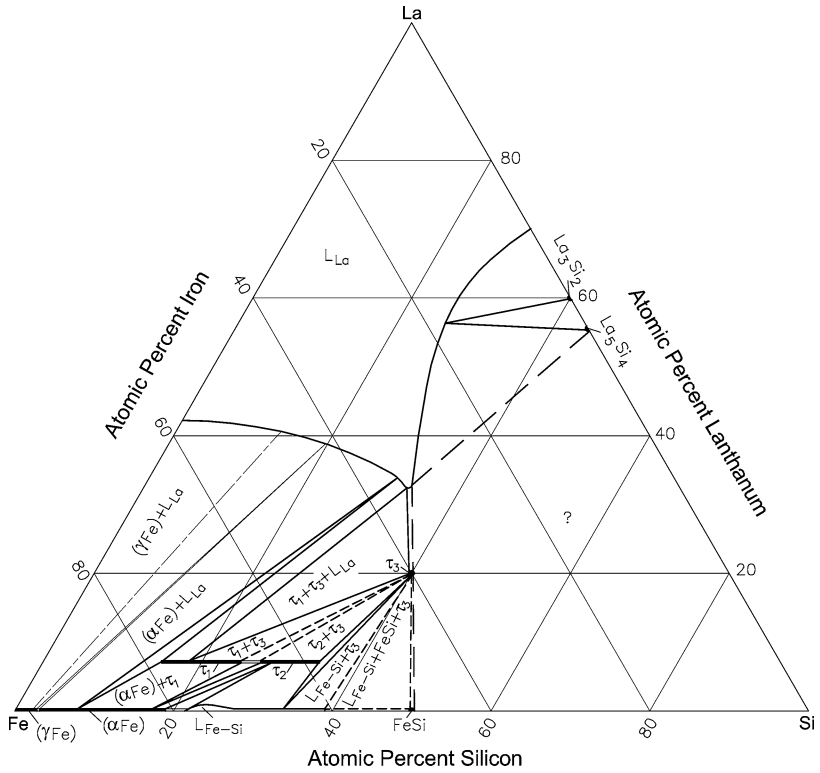


Fig. 1 Fe-La-Si partial isothermal section at 1300 °C [2012Nii]

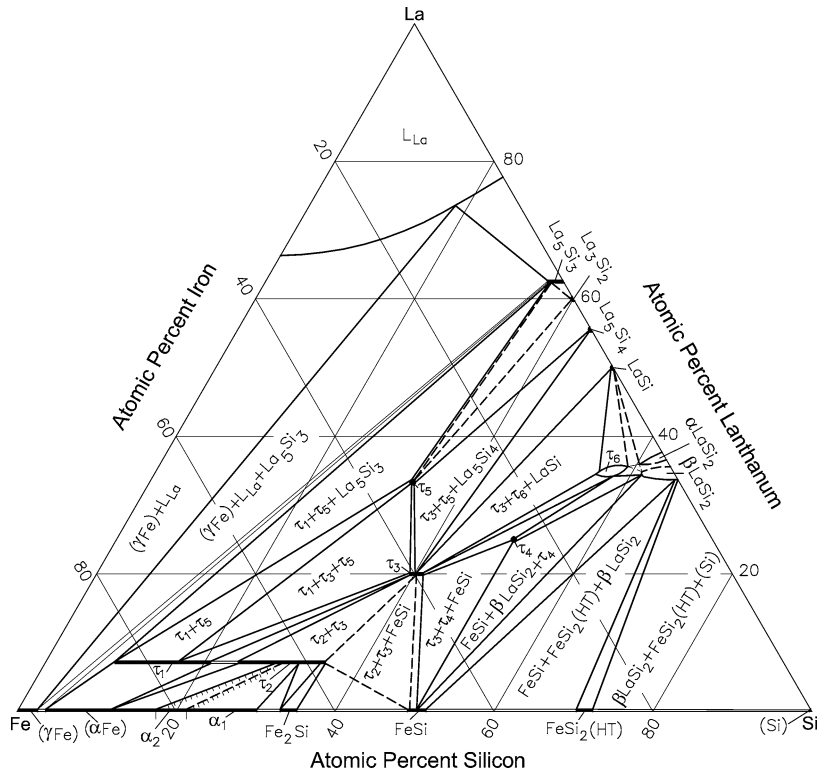


Fig. 2 Fe-La-Si isothermal section at 1100 °C [2012Nii]

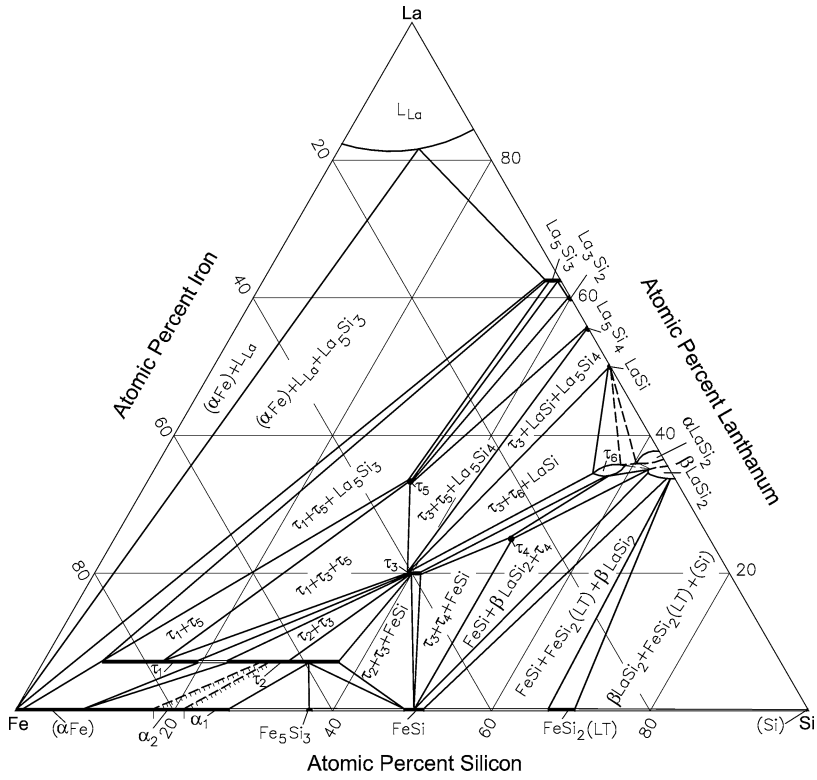


Fig. 3 Fe-La-Si isothermal section at 900 °C [2012Nii]

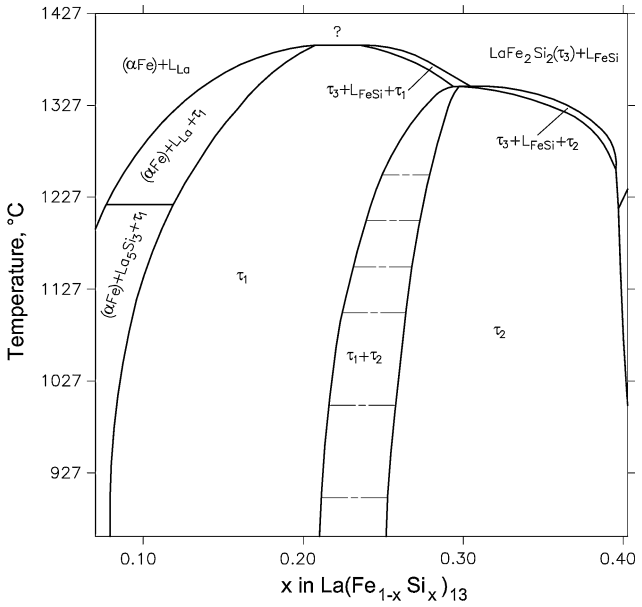


Fig. 4 Fe-La-Si vertical section along the $La(Fe_{1-x}Si_x)_{13}$ line from $x = 0.07$ to $x = 0.40$ [2012Nii]

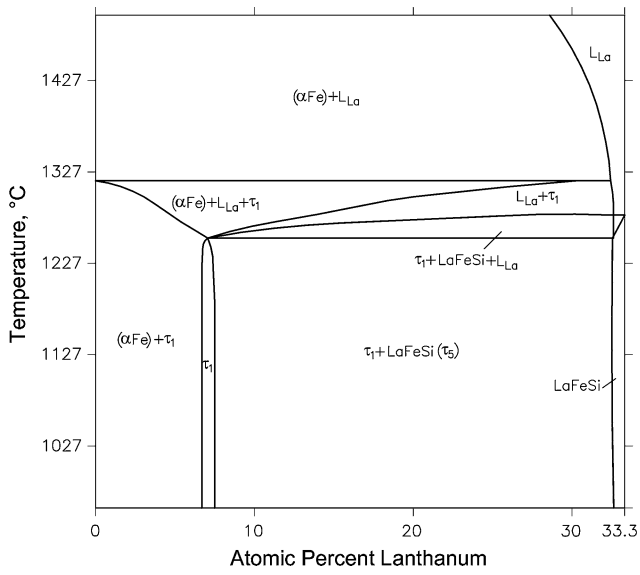


Fig. 5 Fe-La-Si vertical section along the $Fe_{95}Si_5$-LaFeSi join [2012Nii]

appreciable homogeneity range at constant La content. Among the other ternary phases, $LaFe_xSi_{2-x}$ (τ_6) exhibits a small variation in composition.

Two vertical sections were also constructed by [2012Nii] along the $La(Fe_{1-x}Si_x)_{13}$ line from $x = 0.07$ to $x = 0.40$ and along the $Fe_{95}Si_5$-LaFeSi join, as shown in Fig. 4 and 5 respectively. Along the line of 7.1 at.% La (Fig. 4), single-phase regions of τ_1 and τ_2 are present. As the temperature increases, the boundaries of the two-phase ($\tau_1 + \tau_2$) region lean towards the higher Si side. The regions to the left of τ_1

and to the right of τ_2 are three-phase, as the transition across the boundary occurs through a corner of the tie-triangle. [2012Nii] suggested that τ_1 and τ_2 form by ternary peritectic reactions: (αFe) (or $L_{Fe\text{-}Si}$) + L_{La} + $\tau_3 \leftrightarrow \tau_1$ at ~1392 °C and τ_1 + $L_{Fe\text{-}Si}$ + $\tau_3 \leftrightarrow \tau_2$ at 1350 °C. Here, L_{La} and $L_{Fe\text{-}Si}$ stand for La-rich and La-poor liquids. In Fig. 5, the ternary phase τ_1 has adjoining two-phase regions of (αFe) + τ_1 and τ_1 + LaFeSi (τ_5).

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